R16

Code No: 136FD

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, May - 2019
GEOINFORMATICS
(Common to EEE, ME, CSE, MIE)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

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PART - A

(25 Marks)

1.a) What is meant by Focal length of lens in aerial photography? [2]
b) How many minimum numbers of photographs are required for Stereo Photogrammetry? [3]
c) What do you understand by remote sensing? [2]
d) Write the special needs of sensors for geological studies. [3]
e) What do you understand by image histogram? [2]
f) Distinguish between supervised and unsupervised classification. [3]
g) List few Urban applications of GIS. [2]
h) List kind of errors that can occur in GIS. [3]
i) How do you explain GNSS? [2]
j) What do you understand by GALILEO? [3]

PART - B

(50 Marks)

2.a) Discuss about various types of Aerial Photogrammetric techniques.
b) What are the factors effecting aerial photographs? [5+5]

OR

3.a) Explain briefly about digital photogrammetry.
b) Write a short note on Relief Displacement. [5+5]

4. List and explain any two satellites and write down characteristics of satellites. [10]

OR

5.a) Explain about physics of Remote Sensing.
b) Discuss about Remote Sensing satellites and their products. [5+5]

6.a) Explain crop inventory using remote sensing.
b) Explain the remote sensing application in land use and land cover studies. [5+5]

OR

7. Explain the applications of remote sensing in ground water prospects and potential recharge zones. [10]

8.a) Classify data in GIS context and explain spatial data editing.
b) Describe the theoretical frame work and operations involved in Geographic Information System. [5+5]

OR

9.a) Discuss in brief various applications of GIS in civil engineering.
b) What do you understand by spatial data and how are they integrated to make a GIS? [5+5]

10.a) Explain in brief how GPS helps in modern life.
b) Discuss in brief about GPS satellite signals and also explain its importance. [5+5]

OR

11.a) Give the details of the important map projections applicable to Indian regions.
b) Explain in detail the applications of GPS. [5+5]

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